FIN Programs P488–P496

Low-Compute Analytical Research

Persistent spectral cycles, relational time, adaptive memory,
dynamic kernel carriers, topology, and operational identifiability

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Scope. This report executes the nine recommendations from the FIN Theory Compendium that do not require a large Gröbner-basis elimination, a global exact cone optimization, laboratory access, or an external audit. P485, P487, and the global exact phase-face problem are deliberately excluded.

Evidence boundary. Exact algebra, analytic derivations, small dense linear algebra, and synthetic finite records are admitted. No repository-generated record is represented as external physical evidence. New dynamical laws are labelled as supplied models rather than consequences of the frozen kernel.

Runtime. The full numerical batch completed locally in approximately 32 seconds. Ten regression tests completed in 0.18 seconds.

Repository. https://github.com/hyconiek/Fractal-Nadsoliton-Theory

Abstract

Nine low-compute FIN studies are executed. First, a new saturating strict spectral flow

$$\dot{\psi} = (r - g|\psi|^2)\psi - iA\psi$$

is constructed. Every constant-modulus Fourier eigenmode of the strict $\mathbb{Z}_{12}$ generator is an exact persistent rotating state. Five excited modes have strictly negative transverse exponents in the numerical linearization, while mode three has an additional neutral exponent. All have inverse participation ratio 1/12, so the construction gives persistent coherent cycles but not a localized soliton.

The first excited cycle defines a local relational clock with dimensionless period 8.3317982424. Its phase is invariant under simultaneous rescaling of generator and time, proving that the clock cannot select an absolute second. A positive log-parameter flow on the three-pole Stieltjes trace response preserves complete monotonicity and recovers the declared response to 1.31×10^{-15} relative error, but its Jacobian condition number is 3.52×10^{7} and the target/loss remain supplied.

An exact dynamic fractal fixed point is excluded for the declared normalized $C_{12}, C_{24}, C_{48}, C_{96}$ attenuation-envelope contexts: the distinct visible pole counts are $3, 6, 12, 24$, and the finite-size response discrepancies are nonzero and nonmonotone. The identity $A = sI - W$ proves that maximum coupling resonance is exactly minimum Dirichlet cost, whereas maximum temporal frequency reverses this ordering.

An exponent-lift completion automaton is proved to generate both Legacy* and strict formulas exactly for all integer distances and across three carrier sizes. It represents, but does not derive, the parameter transition; the naive linear parameter homotopy becomes a nonnegative-weight passive C_{12} Laplacian only at sampled $t \geq 0.93$. A topology audit proves that the scalar space $\mathbb{C}^{12}$ has no protected winding sectors; a nonvanishing phase field and refinement rule are additional obligations. Finally, coherence-sensitive Fourier measurements raise synthetic four-model classification from 0.5775/0.745 to 1.0 in the declared test, while a dependency-minimal exact checker replays the Dirichlet, Schur, resonance, and Legacy* recurrence identities.

The batch therefore constructs three useful missing mathematical objects—a persistent spectral cycle, a relational clock, and a class-preserving adaptive memory flow—but does not generate a localized nadsoliton, physical time, a learning objective, particle topology, or laboratory evidence.

Status vocabulary. **[PROVEN]** denotes an analytic statement or exact finite replay under declared assumptions; **[STRONG EVIDENCE]** denotes robust numerical or synthetic evidence; **[CONDITIONAL]** denotes a theorem after a supplied dynamical or operational law; **[SPECULATIVE]** denotes an open research interpretation; **[REFUTED]** denotes a scoped counterexample or no-go.

Contents

1 Executive results

Principal advance. A minimal gain–saturation law turns every strict Fourier eigenmode into an exact persistent rotating state. This is the first result in the selected batch that mathematically realizes the "information persists through self-reinforcement" intuition. It is conditional on the new nonlinear law and does not yet realize spatial localization.

Table 1: Programs, new objects, and terminal verdicts.

ID	New typed object	Main result	Status
P488	O192 Saturating Strict Spectral Limit-Cycle Flow	Exact persistent rotating modes; five excited modes strictly transversely damped numerically, one neutral; all delocalized.	[CONDITIONAL] / [STRONG EVIDENCE]
P489	O193 Relational Spectral Phase Clock	Relative phase time exists modulo (8.3318); absolute scale remains a free orbit.	[PROVEN]
P490	O194 Positive Stieltjes Parameter Learning Flow	Positivity and complete monotonicity are preserved; declared response recovered, but ill-conditioned.	[PROVEN] / [STRONG EVIDENCE]
P491	O195 Normalized Dynamic Context Fingerprint	Exact fixed point excluded on the declared four-size family; no monotone convergence evidence.	[PROVEN]
P492	O196 Resonance–Dirichlet Complementarity	Maximum W-coupling equals minimum A-cost; maximum A-frequency is different.	[PROVEN]
P493	O197 Exponent-Lift Kernel Completion Automaton	Legacy* and strict are exact parameter points of one state map; no parameter source is created.	[PROVEN]
P494	O198 Phase-Field Topology Obligation	Scalar kernel states do not possess protected winding sectors.	[PROVEN]
P495	O199 Local Operational Identifiability Atlas	Coherence-sensitive records separate four declared synthetic dynamics; population records do not.	[STRONG EVIDENCE]
P496	O200 Dependency-Minimal Algebraic Replay	Four core identities replayed exactly with standard-library rational arithmetic.	[PROVEN] in replay scope

2 Shared strict finite core

On $V = \mathbb{Z}/12\mathbb{Z}$, let $W_{xx} = 0$ and

$$W_{xy} = \frac{\cos(0.18575d(x,y) + 0.16250)}{1 + d(x,y)^{1.8}}, \qquad A = sI - W,$$

with cyclic distance and

$$s = 1.6603072787660986\ldots.$$

All six sampled off-diagonal weights are positive, so

$$\langle f, Af\rangle = \frac{1}{2}\sum_{x,y} W_{xy}|f_x - f_y|^2 \geq 0.$$

The gap and maximum eigenvalue are

$$\lambda_1 = 0.7541211542070796, \qquad \lambda_{\max} = 2.3421820411463004.$$

This matrix is the fixed input to P488, P489, P490, P492, P494 and P495. P491 declares a separate positive attenuation-envelope refinement family; P493 keeps legacy and strict parameters separated inside one newly supplied state map.

3 P488 — a conditional persistent nadsoliton mechanism

3.1 The new nonlinear object

The frozen strict matrix supports oscillation but does not supply gain, saturation or learning. Introduce the finite complex flow

$$\boxed{\dot{\psi}_x = (r - g|\psi_x|^2)\psi_x - i(A\psi)_x}, \qquad r, g > 0.$$

The real term is uniform gain with local cubic saturation; the skew term is the strict unitary generator. The law is a newly supplied candidate, not a strict consequence.

Theorem 3.1 (Exact strict spectral cycles)**.** *Let $v_m(x) = 12^{-1/2}e^{2\pi imx/12}$ and $Av_m = \lambda_m v_m$. Then*

$$\psi_m(t) = \sqrt{\frac{12r}{g}}\, e^{-i\lambda_m t} v_m$$

is an exact solution. Every site has amplitude $\sqrt{r/g}$.

Proof. Every Fourier coordinate has equal modulus, so

$$g|\psi_m(x)|^2 = g\frac{12r/g}{12} = r.$$

Gain and saturation cancel pointwise, leaving $\dot{\psi}_m = -iA\psi_m = -i\lambda_m\psi_m$. □

With $r = 0.2$, $g = 1$, exact equation residuals are below 3.0×10^{-15}. The inverse participation ratio is

$$\sum_x |v_m(x)|^4 = \frac{1}{12}$$

for every mode. Thus the solutions persist but are maximally delocalized on the carrier.

3.2 Linearized stability audit

The real 24×24 co-rotating Jacobian was diagonalized for modes 0 through 6. One zero exponent is required by global phase symmetry. After removing it, the largest real parts are:

Mode	Frequency	Period	Largest transverse real part
0	(0)	—	(-0.2000000)
1	(0.7541212)	(8.3317982)	(-0.0029812)
2	(1.5770495)	(3.9841395)	(-0.0065292)
3	1.9614069	3.2034074	1.10×10^{-16}
4	(2.1995688)	(2.8565531)	(-0.0124871)
5	(2.2986063)	(2.7334761)	(-0.0019318)
6	(2.3421820)	(2.6826204)	(-0.0048048)

Modes (1,2,4,5,6) are strictly damped transversely in floating linearization. Mode (3) has an additional neutral direction and is not promoted to asymptotic stability. No positive exponent was detected. These are **[STRONG EVIDENCE]**, not interval-certified Floquet theorems.

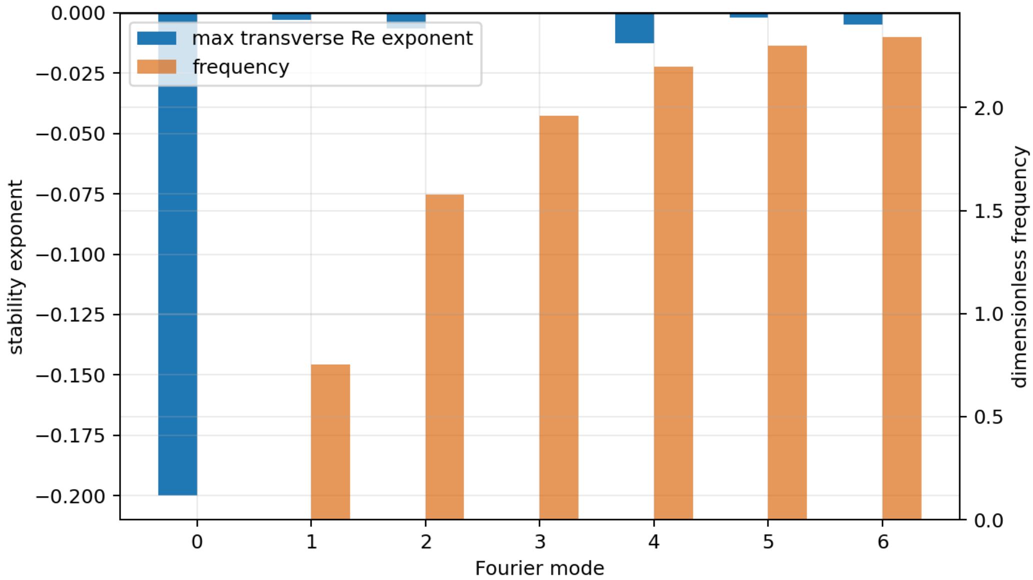

Figure 1: Frequencies and largest transverse real exponents for the supplied saturating flow. Exact persistence and numerical stability are different claim layers.

Verdict. The batch constructs persistent self-reinforced information cycles. It does not establish a localized travelling soliton, a continuum limit, or the physical law that selects r and g.

4 P489 — relational time from a persistent frequency

Select mode $m = 1$. Its recorded complex coefficient has phase

$$z(t) = z(0)e^{-i\lambda_1 t}, \qquad \lambda_1 = 0.75412115420708.$$

On an interval shorter than

$$T_1 = \frac{2\pi}{\lambda_1} = 8.3317982424005,$$

an unwrapped phase defines t relative to a chosen origin. Beyond one period, a single phase aliases times modulo T_1.

Proposition 4.1 (Affine scale obstruction)**.** *For every (c>0), the replacements*

$$A \mapsto A/c, \qquad t \mapsto ct$$

leave e^{-itA} and every phase record unchanged. Therefore the spectral clock cannot choose an absolute time unit.

The executed check used 41 synthetic phase samples with noise standard deviation 0.005 radian. Linear unwrapping estimated λ_1 with relative error 3.10×10^{-4}. The scale-orbit residuals for $c = 0.1, 1, 10$ were below 9.0×10^{-16}.

If phase error is ϵ_ϕ and relative frequency error is $\epsilon_\lambda < 1$, a local first-order timing bound over horizon T is

$$|\hat{t} - t| \lesssim \frac{\epsilon_\phi/\lambda_1 + T\epsilon_\lambda}{1 - \epsilon_\lambda}.$$

For $\epsilon_\phi = 0.01$, $\epsilon_\lambda = 10^{-3}$, and $T = 0.8T_1$, this gives 0.01995 dimensionless time units.

Verdict. The user's frequency-as-time intuition survives as a relational clock theorem. It does not create a second, select the clock mode, or prove that the physical universe uses this cycle.

5 P490 — adaptive learning in Stieltjes memory space

5.1 Context response

Split the strict C_{12} carrier into even visible and odd hidden coordinates and regularize the hidden block by $0.15I$. The normalized trace self-energy is

$$\sigma(z) = \frac{1}{6}\,\mathrm{Tr}\Big[B(zI + C)^{-1}B^T\Big] = \sum_{j=1}^{3} \frac{r_j}{z + c_j},$$

with positive poles and residues

j	1	2	3
c_j	1.3210910206	1.6763637131	2.0383091819
r_j	0.2285756964	0.1987861899	0.0322941935.

Theorem 5.1 (Class-preserving adaptive parameterization)**.** *Write $c_j = e^{b_j}$, $r_j = e^{a_j}$, and update a, b by any finite gradient step that remains finite. Then $c_j, r_j > 0$, and*

$$(-1)^k \sigma^{(k)}(z) = k! \sum_j \frac{r_j}{(z + c_j)^{k+1}} > 0$$

for (z>0). The learned response remains in the rational Stieltjes class.

Twelve deterministic-seed random starts recovered the 48-point trace response below 10^{-6}; the best maximum relative error was 1.31×10^{-15}. Positivity margins through derivative order four remained strictly positive. The relative Jacobian condition number was 3.52×10^{7}, warning that accurate response does not imply robust pole identification.

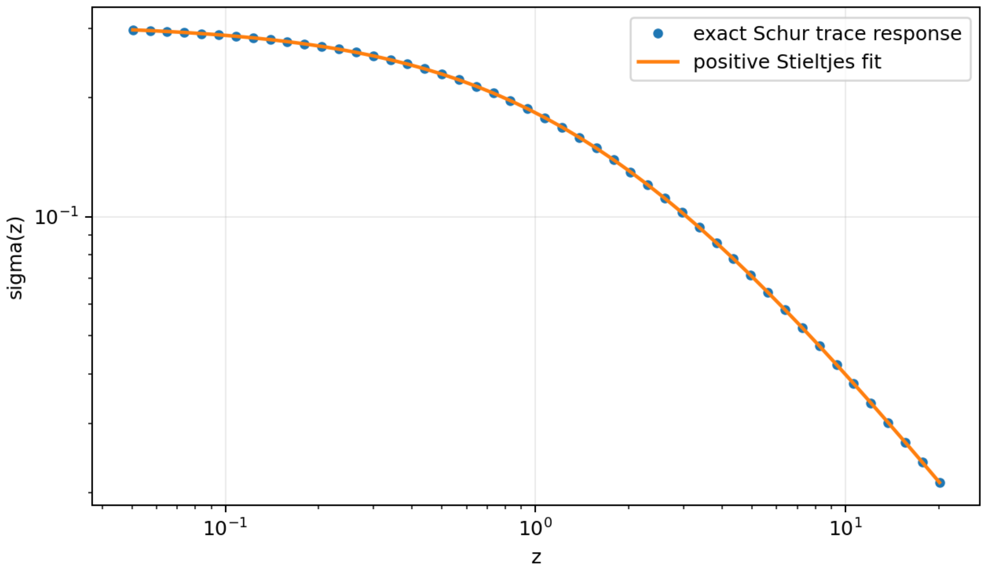

Figure 2: Exact strict Schur trace response and the learned positive three-pole Stieltjes representation.

Verdict. FIN now has a mathematically natural adaptive law in memory-function space. The construction does not internally generate its training target, loss, observations, or learning rate.

6 P491 — dynamic fractal fixed-point test

To avoid importing the strict oscillatory phase outside its canonical C_{12} scope, this programme declares the positive attenuation family

$$h(d) = \frac{1}{1 + d^{1.8}}$$

on C_n, $n = 12, 24, 48, 96$. Even vertices are retained, odd vertices hidden, and $0.1I$ regularizes the hidden block. Define the scale-normalized trace fingerprint

$$F_n(u) = \frac{\mathrm{Tr}[B(u\, c_n I + C)^{-1} B^T]}{\mathrm{Tr}[B(c_n I + C)^{-1} B^T]},$$

where c_n is the median hidden eigenvalue.

The distinct visible pole counts were

$$3, \quad 6, \quad 12, \quad 24,$$

for $C_{12}, C_{24}, C_{48}, C_{96}$. Successive maximum fingerprint differences on the seven-point u-grid were

$$0.0059889, \qquad 0.0129334, \qquad 0.0081270.$$

Proposition 6.1 (Declared finite-family obstruction)**.** *Two rational Stieltjes functions with positive noncancelling residues can be identical only if their pole sets and residues agree. The declared fingerprints have different visible pole counts and nonzero response differences. Hence they are not an exact dynamic fixed point of this normalization.*

The discrepancies do not decrease monotonically, so these four sizes do not supply evidence of numerical convergence either. They do not prove that no different refinement, normalization or infinite-dimensional fixed point can exist.

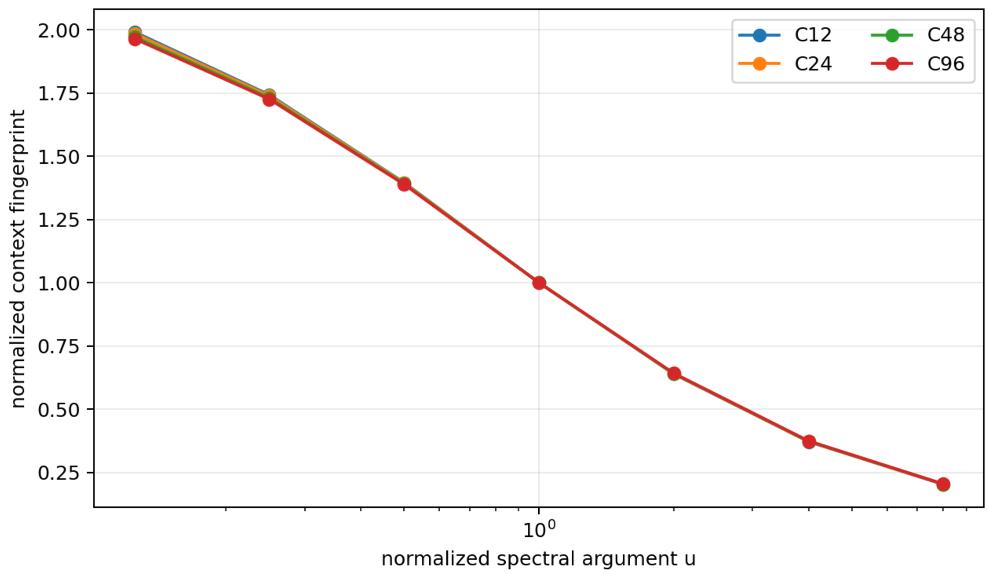

Figure 3: Normalized dynamic context fingerprints. Their proximity is approximate; pole counts exclude exact equality in the declared family.

7 P492 — what “maximum resonance” means mathematically

Theorem 7.1 (Resonance–Dirichlet complementarity)**.** *For every nonzero $x \in \mathbb{C}^{12}$,*

$$\frac{\langle x, Wx\rangle}{\langle x, x\rangle} = s - \frac{\langle x, Ax\rangle}{\langle x, x\rangle}.$$

Consequently maximizing coupling resonance of W is exactly minimizing Dirichlet cost of A on the same constraint set.

The random numerical replay residual was 8.89×10^{-16}; the standard-library rational checker verifies the identity exactly on 80 instances. On the mean-zero subspace,

$$\max R_W = 0.9061861245590195, \qquad \min R_A = 0.7541211542070796,$$

and their sum is s.

"Highest temporal resonance" can instead mean maximum A-frequency. The paired values are

$$\lambda_{\max}(A) = 2.3421820411463004, \qquad \lambda_{\min}(W) = -0.6818747623802017.$$

This criterion reverses the coupling ordering.

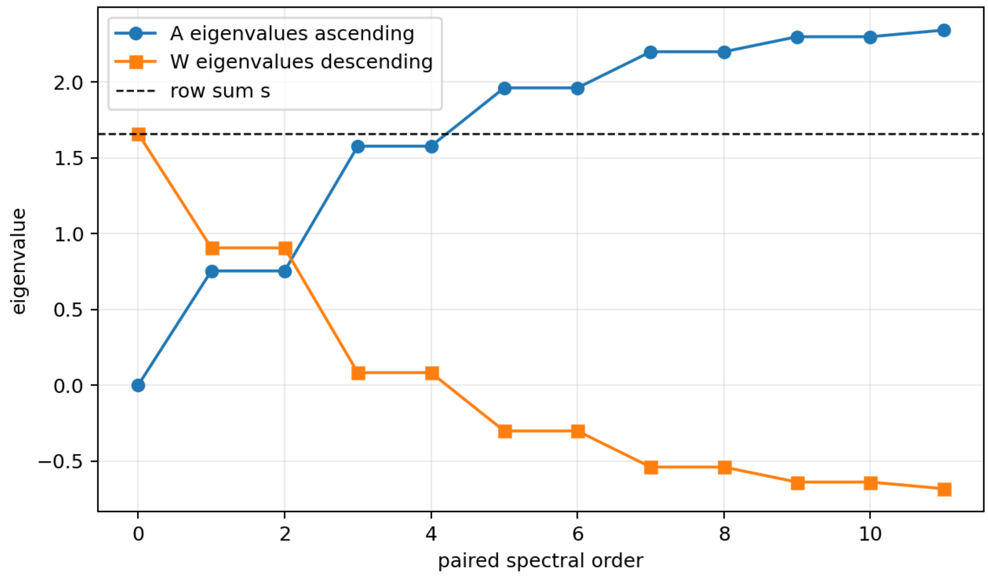

Figure 4: Paired spectral ordering. The sum of corresponding W and A eigenvalues is the row sum s.

Verdict. The founding phrase needs a declared observable. Maximum coupling is not an alternative to minimum Dirichlet cost in the current strict model; it is the same variational statement. Sustained excitation is supplied by P488's gain–saturation law, not by spectral ordering alone.

8 P493 — a typed dynamic carrier for Legacy* and strict

8.1 Exponent-lift automaton

For parameters $p = (a, \beta, \eta, \omega, \phi)$, define

$$q_{d+1} = q_d + 1, \qquad c_{d+1} = \frac{c_d}{1 + \beta[(q_d+1)^\eta - q_d^\eta]c_d}, \qquad \theta_{d+1} = \theta_d + \omega,$$

with $q_0 = 0$, $c_0 = 1$, $\theta_0 = \phi$, and output

$$K_d = ac_d \cos\theta_d.$$

Theorem 8.1 (Exact exponent-lift representation)**.** *For every integer $d \geq 0$,*

$$c_d = \frac{1}{1 + \beta d^\eta}, \qquad K_d = a\frac{\cos(\omega d + \phi)}{1 + \beta d^\eta}.$$

Proof. The result is true at $d = 0$. If it holds at d, then

$$c_{d+1} = \frac{1}{1 + \beta d^\eta + \beta[(d+1)^\eta - d^\eta]} = \frac{1}{1 + \beta(d+1)^\eta}.$$

The phase recurrence is immediate. □

The Legacy* point is

$$(a, \beta, \eta, \omega, \phi) = (4 \ln 2, 0.01, 1, \pi/4, \pi/6),$$

while strict is

$$(1, 1, 1.8, 0.18575, 0.16250).$$

The automaton matched both closed forms through distance 48 to below floating tolerance and transferred identically on C_{12}, C_{16}, C_{24}.

This is representation, not derivation: every endpoint parameter is supplied. A linear homotopy between parameter tuples retains one or more negative radial weights until the sampled parameter $t = 0.93$. Therefore most of that simple path is not a nonnegative-weight passive graph Laplacian. No physical legacy role transfers through this automaton.

Verdict. The construction identifies a compact dynamic state space containing both kernel classes and isolates the missing problem as a parameter-source law. It does not solve that problem.

9 P494 — topology and the particle-defect boundary

Fourier phase patterns on the cycle have discrete winding m for increments strictly inside $(-\pi, \pi)$. Their strict Dirichlet energies are the spectral values:

$$E_{\pm 1} = 0.7541212, \quad E_{\pm 2} = 1.5770495, \quad E_{\pm 3} = 1.9614069,$$
$$E_{\pm 4} = 2.1995688, \quad E_{\pm 5} = 2.2986063.$$

The Nyquist mode $m = 6$ lies exactly at edge increment π, so its signed winding depends on branch convention and is marked ambiguous.

Theorem 9.1 (Scalar-kernel topology obstruction)**.** *The unconstrained scalar configuration space $\mathbb{C}^{12}$ is contractible. Hence it has no disconnected winding sectors. A path from the $m = 1$ Fourier field to the constant field can pass through the zero field with finite strict Dirichlet energy. Protected winding requires at least:*

1. *a nonvanishing S^1-valued phase field or equivalent target constraint;*
2. *a declared interpolation on cycle edges;*
3. *a refinement/continuum rule preserving the homotopy class.*

The finite kernel can assign energy to a supplied winding pattern; it cannot create the target space or identify that pattern with an electron, muon, tau, or any other particle.

10 P495 — a small operational identifiability atlas

Four declared finite models were compared:

- strict unitary $\rho \mapsto e^{-itA} \rho e^{itA}$;
- strict Markov populations $p \mapsto e^{-tA} p$;
- vertex-dephasing Lindblad dynamics with $\gamma = 0.6$;

- a completely positive synthetic coherence-revival family

$$\rho(t) = q(t)U_t\rho U_t^\dagger + [1 - q(t)]\,\mathrm{diag}(U_t\rho U_t^\dagger), \quad q(t) = 0.45 + 0.40\cos(2.8t).$$

Six preparations, three times and vertex/Fourier measurements were available. Multinomial records used (120) shots per context and (100) trials per true model.

Design	Contexts	Minimum mean TV	Classification accuracy
D1: one time, basis, vertex	3	1.16×10^{-18}	0.5775
D2: multitime, all preparations, vertex	18	2.05×10^{-17}	0.7450
D3: multitime, all preparations, vertex+Fourier	36	(0.0487983)	(1.0000)

D1 and D2 contain an exactly/near-exactly indistinguishable pair in their declared population records: dephasing a coherent state does not change its instantaneous vertex diagonal. Fourier measurement exposes the missing coherence and separates every model in this synthetic dataset.

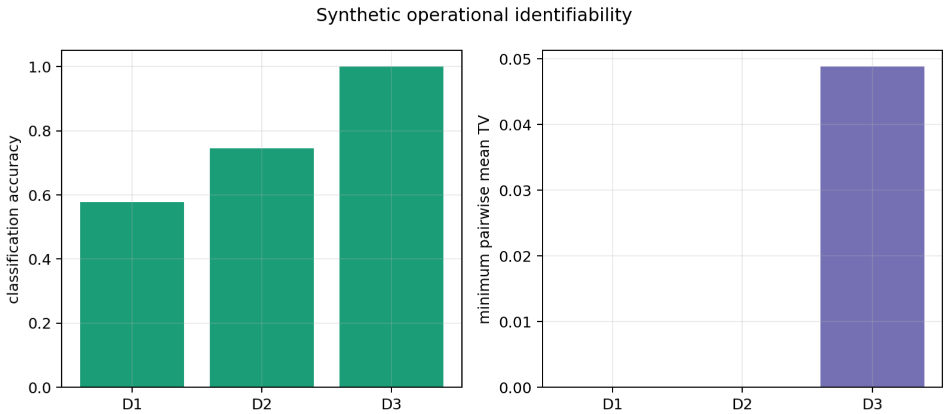

Figure 5: Operational resources matter more than additional population samples for the declared unitary/revival ambiguity.

Verdict. The result supports the laboratory package's demand for informationally complete preparations and instruments. Perfect synthetic classification is not a laboratory prediction, hardware tolerance theorem, or external validation.

11 P496 — dependency-minimal exact replay

The standard-library checker uses exact rational arithmetic and no NumPy, SciPy, SymPy or network resource. It passed:

- 80 exact regular-circulant Dirichlet identities;
- 80 exact Rayleigh complementarity identities;
- 60 exact positive rational Green–Schur block identities;
- 25 exact Legacy* $\eta = 1$, $\beta = 1/100$ automaton steps.

The checker strengthens reproducibility and rejects accidental floating-only algebra. It is not a proof-assistant kernel, does not certify the transcendental strict decimals, and does not replace the analytic proofs stated above.

12 Integrated consequences for FIN

12.1 What has genuinely improved

1. **Persistence now has a concrete model.** The P488 law is the smallest executed mechanism in this batch combining positive self-reinforcement, saturation, and strict spectral phase evolution.
2. **Frequency can carry relative time.** P489 converts the user's intuition into a theorem with an explicit alias period and error ledger.
3. **Learning can occur in the right object space.** P490 adapts a Green–Schur memory response while preserving the Stieltjes cone.
4. **Legacy and strict share an exact dynamic syntax.** P493 supplies a state recurrence producing both formula classes at different parameters.
5. **The missing topology is precisely typed.** P494 identifies the nonvanishing phase field and refinement rule that particle-defect claims require.
6. **The observer needs coherence access.** P495 shows why population-only records can leave unitary and revival descriptions indistinguishable.

12.2 What has not improved

- P488's modes are not localized solitons.
- The gain, saturation, learning loss and clock choice are additional laws.
- The absolute time, length and action scales remain absent.
- No exact fractal dynamic fixed point was found in the declared family.
- P493 does not derive strict parameters from legacy and transfers no old physical role.
- No particle, mass, gauge field, gravity or laboratory record is generated.
- The selector obstruction QW-2191 is untouched.

Updated interpretation. The most promising low-cost continuation of FIN is no longer another static kernel fit. It is a nonlinear spectral-memory theory: strict supplies the conservative phase geometry, a separately justified gain–saturation law supplies persistence, and Stieltjes–Schur state variables supply adaptive memory. Localization, parameter provenance, operational calibration, and topology remain the decisive missing theorems.

13 Recommended next low-compute studies P497–P506

These recommendations remain local and bounded. They do not reopen the high-cost exact-algebra jobs.

Table 2: Next analytical/small-compute programmes.

Program	Objective	Acceptance criterion	Priority
P497	Localized nonlinear strict states	Continue stationary solutions from a single-site anti-continuum limit; measure IPR and prove/refute persistence under strict coupling.	10/10
P498	Certified P488 stability	Enclose every nonzero real Jacobian exponent by interval arithmetic, separating the neutral $m = 3$ direction.	8/10

Program	Objective	Acceptance criterion	Priority
P499	Two-frequency clock atlas	Prove periodicity versus injectivity from rational/irrational frequency ratios; quantify finite-precision recurrence.	8/10
P500	Stieltjes identifiability bounds	Convert the 3.52×10^7 condition number into rigorous noise-to-pole uncertainty bounds.	9/10
P501	Context-family limit test	Extend P491 to C_{192}, C_{384} and derive an asymptotic integral fingerprint before claiming or rejecting convergence.	7/10
P502	Exact passivity boundary	Interval-isolate the first P493 homotopy parameter with all six radial weights nonnegative and audit PSD separately.	7/10
P503	Nonvanishing phase refinement	Add an S^1-valued field and prove a discrete winding preservation theorem under bounded updates.	9/10
P504	Minimal coherence tester	Remove preparations/measurements from D3 until the smallest still-separating synthetic tester is certified.	8/10
P505	Formal theorem interface	Translate P488 existence, P489 scale orbit, P492 complementarity and P493 induction into a proof-assistant-neutral theorem dependency file.	7/10
P506	Parameter-source identifiability no-go	Determine which observations could identify a unique trajectory in P493 parameter space without inserting the strict endpoint.	10/10

P497 is scientifically the most important: localization is the missing property separating a persistent spectral oscillator from a soliton. P500 and P506 test whether the new adaptive/dynamic objects can ever become predictive rather than fitted.

14 Reproducibility

Primary artifacts:

- `fin_programs_488_496_low_compute.py`;
- `fin_programs_488_496_exact_checker.py`;
- `test_fin_programs_488_496.py`;
- `FIN_Programs_488_496_Results.json`;
- `FIN_Programs_488_496_Exact_Check.json`;
- `FIN_Programs_488_496_Summary.csv`;
- `FIN_Programs_488_496_Figures/`.

Local commands:

```
MPLCONFIGDIR=/tmp/fin-mpl-1048 \
  python3 fin_programs_488_496_low_compute.py
python3 -m unittest -v test_fin_programs_488_496.py
pdflatex -interaction=nonstopmode -halt-on-error \
  FIN_Programs_488_496_Low_Compute_Analytical_Report_EN.tex
```

The full run used one deterministic seed, dense matrices of size at most 144×144, refinement matrices no larger than 96×96, and small nonlinear least squares. It is designed for an ordinary local computer.

15 Final verdict and nonclaims

The low-compute route was scientifically productive. It did not solve the most expensive exact-algebra problems, yet it constructed objects closer to the founding FIN idea than another static interpolation would have done. The strict spectrum can support persistent self-reinforced cycles after a minimal saturating law is supplied. Their frequency gives a rigorous relational clock. Hidden-state response admits a positivity-preserving learning law. Legacy* and strict fit one exact dynamic recurrence syntax. These are real mathematical advances within declared models.

The same batch sharply limits interpretation. The cycles are not localized. The clock has no absolute unit. The adaptive response is ill-conditioned and trained on supplied data. The dynamic kernel automaton imports all endpoint parameters. Scalar states have no protected particle winding. Population measurements alone do not determine the temporal model.

No result here discharges QW-2191, sources orientation, produces dimensional standards, completes the legacy-to-strict bridge, transfers a legacy physical role, supplies independent custody or laboratory evidence, derives particles, masses, the Standard Model, gravity, L_{total}, or a Theory of Everything.